Data Storage in Android

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Outline

1 Intoduction

2 SharedPreferences

3 Files

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Types

- `SharedPreferences`: Store private, primitive data in key-value pairs.
- App-specific storage (Internal storage): Store files that are meant for your app's use only.
- Shared storage (External storage): Store files that your app intends to share with other apps
- Databases: Store structured data in a private database using the Room persistence library.

SharedPreferences

- store a small collection of values
- use key value pairs
- only primitive types

Write SharedPreferences

```
val sharedPref = activity?.getPreferences(Context.MODE_PRIVATE)
    ?: return
with (sharedPref.edit()) {
    putInt(getString(R.string.saved_high_score_key), newHighScore
        )
    apply()
}
```

Read SharedPreferences

```
val sharedPref = activity?.getPreferences(Context.MODE_PRIVATE)
    ?: return
val defaultValue = resources.getInteger(R.integer.
    saved_high_score_default_key)
val highScore = sharedPref.getInt(getString(R.string.
    saved_high_score_key), defaultValue)
```

Advanced

- specific preference file

```
val sharedPref = activity?.getSharedPreferences(getString(R.
    string.preference_file_key), Context.MODE_PRIVATE)
```

- Encrypted shared prefenreces: `EncryptedSharedPreferences`, to store sensitive data
- DataStore: store any typed object not only the primitive types

Types

- Internal storage directories: These directories include both a dedicated location for storing persistent files, and another location for storing cache data.
- External storage directories: These directories include both a dedicated location for storing persistent files, and another location for storing cache data. (Needs a permission)

Access files

```
val filename = "myfile"
val fileContents = "Hello world!"
context.openFileOutput(filename, Context.MODE_PRIVATE).use {
  it.write(fileContents.toByteArray())
}
```

Types

- SQLiteOpenHelper
- Room
- LivingRoom